

# Abhijeet Sandip (Abhi) Pachpute

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## Professional Experience

### AI Engineer

July 2025 - Present

*Paxel AI (pharma sales intelligence startup)*

Salt Lake City, UT, USA

- Built an intelligence catalog over a multi-million-row pharmaceutical sales dataset in Aurora PostgreSQL, mapping 8 business domains into one curated, cited knowledge layer that is provider-agnostic and portable, so any service can build AI-native features on the data without re-modeling it
- Built the production AI assistant on it, a governed agent that turns plain-English rep questions into safe SQL over their tenant's data, on AWS ECS Fargate (dynamic task allocation) with Aurora behind RDS Proxy; streams answers live as the model generates them, at about 90% accuracy and a ~5s median response
- Designed the governed data and safety layer: 17 Postgres views that enforce tenant and per-rep row scoping in the database, plus a SQL parse gate that permits only a single SELECT over an allowlist, so a model-written query can never reach a base table or issue a write

### AI Software Engineer Intern

May 2025 - August 2025

*AVI Human Services*

Logan, UT (Remote), USA

- Built fault-tolerant REST APIs with Redis caching and tuned MySQL indexes, holding median response times under 200ms under concurrent load across a 50,000+ record dataset
- Added automated alerting and graceful degradation so queries stayed available during partial failures, and shipped a real-time analytics platform (React, Node, TypeScript, Docker on AWS) in an iterative delivery cycle
- Designed a LangChain pipeline with the Gemini API and vector embeddings for automated content generation

### Software Developer Intern

July 2023 - February 2024

*eWarranty Solutions*

Pune, India

- Built a QR-code warranty verification system (Java, Kotlin, Spring Boot, RESTful APIs) for 45,000+ products, cutting manual errors by 30%
- Modeled the analytics pipeline as an async computation graph with Java CompletableFuture chains and a read-optimized MySQL schema, decoupling query latency from write throughput at scale

### IT Systems & Security Intern

June 2025 - May 2026

*University of Utah*

Salt Lake City, UT, USA

- Automated deployment and security workflows with Python across Linux, macOS, and Windows, and resolved system- and network-level issues (TCP/IP, identity systems) with root-cause analysis across a \$650M+ research infrastructure

## Projects

### SynapticaAI - LLM Gateway & API Infrastructure | [GitHub](#)

- Built an OpenAI-compatible API gateway with a configurable routing layer and semantic caching across providers, cutting redundant calls and tail latency
- Instrumented request-level observability (latency, error rate, cache-hit rate) with a live dashboard and CI-gated evals that block regressions before deploy

### Pact - Privacy-first Encrypted Video Calling | [Live Demo](#)

- Built and deployed real-time, end-to-end encrypted audio/video over WebRTC with sub-second peer connection setup, using Cloudflare TURN relay and Supabase for signaling, presence, and session state
- Owned the full build solo, from WebRTC connection handling and encryption to UI and production deployment at [pact.abhijeetpachpute.com](https://pact.abhijeetpachpute.com)

### ClipSync - Self-hosted Multi-device Clipboard | [Live Demo](#)

- Built real-time cross-device sync with Next.js and Supabase, featuring row-level security and offline-tolerant state (Zustand), deployed on Vercel

## Technical Skills

- **Languages:** Python, TypeScript, JavaScript (ES6+), Java, C++, C, Kotlin, Bash/Shell, SQL
- **Backend & Distributed Systems:** REST API design, microservices, async processing, concurrency, caching (Redis), Spring Boot, Node.js; Aurora PostgreSQL, MySQL, MongoDB, Neo4j
- **Cloud & Infrastructure:** AWS (Lambda, EKS, ECR, S3, EC2), Docker, Kubernetes, CI/CD, Linux/Unix, Git/GitHub
- **AI / ML:** LangGraph, LangChain, multi-agent RAG, vector embeddings, OpenAI / Gemini / Claude APIs, HuggingFace

## Research & Recognition

- 2 IEEE publications and 3 filed patents (IP India) in generative models and computer vision. Smart India Hackathon 2022 Finalist. Links on portfolio.

## Education

**M.S. Computer Science** - University of Utah (GPA 3.7/4.0)

August 2024 - May 2026

**B.E. Computer Engineering** - Savitribai Phule Pune University (GPA 8.7/10)

July 2019 - May 2023